

A Small-Parameter Obstruction to the Proposed Mixed Steklov–Dirichlet Two-Term Bound

Run fa_banach_001

2026-08-12

Status. Candidate counterexample, likely valid, to Open Question 1.8 exactly as printed in arXiv:1712.00753. The conclusion concerns the all- $z > 0$ reading used by the surrounding estimates.

1 Source question

Hassannezhad and Laptev study the mixed Steklov–Dirichlet eigenvalues η_j of a bounded domain $\Omega \subset \mathbb{R}^n$ whose boundary is split into a free part $\mathcal{F}$ and a Dirichlet part. For $\gamma > 0$ the Riesz mean is

$$R_\gamma(z) = \sum_j (z - \eta_j)_+^\gamma, \quad z > 0.$$

On page 6, Open Question 1.8 asks whether there is a positive constant $C(\alpha, \beta)$, depending only on the two corner angles, such that

$$R_\gamma(z) \leq C_{n,\gamma} |\mathcal{F}| z^{n+\gamma-1} - C(\alpha, \beta) z^{n+\gamma-2}. \quad (1)$$

No lower cutoff in z and no remainder term is stated [1].

One can apply the Riesz iteration to find bounds on $R_\gamma(z, \mathcal{D}_D)$.

Open Question 1.8. *It is an intriguing question if we can get a two-term upper bound with a negative second term depending only on α and β . One can ask if there exist a positive constant $C(\alpha, \beta)$ such that*

$$\sum_j (z - \eta_j)_+^\gamma \leq C_{n,\gamma} |\mathcal{F}| z^{n+\gamma-1} - C(\alpha, \beta) z^{n+\gamma-2}.$$

2 Idea of the proof

The Riesz mean can never be negative, because it is a sum of nonnegative terms. But the suggested two-term polynomial is necessarily negative near $z = 0$: the negative term has one lower power of z than the positive Weyl term. Factoring out the common positive power makes the contradiction immediate.

3 Counterexample theorem

Theorem 1. *Fix $n \geq 2$, $\gamma > 0$, a domain with $|\mathcal{F}| > 0$, and any proposed constant $C(\alpha, \beta) > 0$. Then inequality (1) fails for some $z > 0$. Consequently, no positive $C(\alpha, \beta)$ can make the inequality in Open Question 1.8 hold for every $z > 0$.*

Proof. Put $A = C_{n,\gamma}|\mathcal{F}| > 0$ and $C = C(\alpha, \beta) > 0$. The right-hand side of (1) factors as

$$z^{n+\gamma-2}(Az - C).$$

Choose any

$$0 < z < \frac{C}{A}.$$

Because $n \geq 2$ and $\gamma > 0$, the factor $z^{n+\gamma-2}$ is positive, while $Az - C < 0$. Hence the proposed right-hand side is strictly negative. On the other hand,

$$R_\gamma(z) = \sum_j (z - \eta_j)_+^\gamma \geq 0.$$

Thus (1) would require a nonnegative number to be at most a strictly negative number, a contradiction. $\square$

The argument does not require a specially chosen domain or spectrum. In particular, if z is also taken below the first mixed Steklov–Dirichlet eigenvalue, then $R_\gamma(z) = 0$ exactly.

4 Deepening and the necessary repair

The obstruction shows that a global estimate with a pure negative second term is impossible. Any corrected version must include at least one of the following: a lower threshold $z \geq z_0$, a nonnegative lower-order remainder, or a positive-part truncation of the polynomial right-hand side.

The source’s immediately preceding planar estimate already displays the relevant repair. For right-angle John domains, Theorem 1.7 gives

$$R_1(z) \leq \frac{|\mathcal{F}|}{2\pi} z^2 - \frac{1}{2} z + \frac{\pi}{2|\mathcal{F}|}.$$

The final positive constant prevents the right-hand side from becoming negative near the origin. Removing it is not a harmless sharpening at small z .

Remark 1. *The argument does not decide the charitable asymptotic reformulation: for a fixed domain, does there exist z_0 such that a bound of the displayed form holds for all $z \geq z_0$? Nor does it determine an optimal explicit positive remainder. Those are distinct repaired questions.*

5 Verification and novelty check

The source formula and its lack of a lower cutoff were checked in the final arXiv PDF, page 6, Open Question 1.8. The script `code/verify_sign.py` chooses $z = C/(2A)$ and verifies that the proposed right side is negative while the universal lower bound on the Riesz mean is zero.

A bounded search on 2026-08-12 used the exact question wording, the source arXiv id, and combinations of *mixed Steklov–Dirichlet*, *Riesz mean*, and *negative second term*. It found the source and adjacent later work on mixed Steklov eigenvalues, but no paper pointing out or repairing this small- z obstruction. Novelty confidence is moderate; mathematical-validity confidence for the literal quantifier is high.

References

- [1] A. Hassannezhad and A. Laptev, *Eigenvalue bounds of mixed Steklov problems*, arXiv:1712.00753.